# Data Loading, Transformation, Modeling, and KPI Setup

## 1. Introduction to the Data Scenario

In many real-world BI projects, data related to a business process comes in multiple files and multiple formats. In this case study, the shipping and order details for a logistics company *Serveto* are distributed across different CSV files for the years 2018–2021, and another dataset is supplied in an Excel file named **Shipping.xlsx**.

The goal is to **combine**, **clean**, and **model** these datasets so that we can build a dashboard showing shipping performance insights. As discussed, the final dashboard focuses on the **year 2021**, but the backend needs a clean combined dataset.

Power BI provides multiple options for combining files—primarily **Append Queries** and **Merge Queries**. Append is used when tables have similar structure; merge is used to join tables based on a lookup key. In this exercise, since the two CSV files share identical column structures (same headers and types), **Append Queries** is the correct choice.

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## 2. Importing the CSV Data Files

### 2.1. Importing Order Data for 2018–2019

The first dataset consists of orders from **2018 and 2019**, stored in a single CSV file. The steps include:

- Click **Home → New Source → Text/CSV**
- Browse the file for 2018–2019.
- Power BI opens a **preview** window and automatically identifies column delimiters and column types.
- Click **Transform Data** to bring the file into the Power Query Editor.



### 2.2. Importing Order Data for 2020–2021

Similarly:

- Click **New Source → Text/CSV**
- Select the CSV containing **2020 and 2021** data.
- Click **Transform Data**.

Now both CSVs are visible as **separate queries** in the Power Query Editor.

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## 3. Appending the CSV Tables

Since:

- Both CSV files have **the same column names**,
- And have **identical data types**,

They can be appended into one dataset using:

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### 3.1. Append Queries Option

Steps:

1. In Power Query, go to the **Home** tab.
2. Click the dropdown under **Append Queries**.
3. Choose **Append Queries as New** (because we want a new table).
4. Power BI opens a window with options to append:
    - 2 tables
    - 3 tables
    - More (if using automatic folder append)
5. Choose the two imported CSV tables.

After appending, Power BI creates a **new query** named something like:

**Append1** or **Appended Table** (name can be changed).

This table now contains **combined data from 2018–2021**.

### 3.2. Verification After Append

Once created:

- Preview the new table in Power Query.
- Scroll through columns to ensure:
    - Column headers are consistent
    - Data types match expectations
    - No null values or mismatched types

This appended table is crucial because it will serve as the **fact table** for order analytics.

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## 4. Adding the Excel Dataset (Shipping Details)

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After appending CSVs, the next dataset (shipping information) is loaded:

### 4.1. Importing the Excel File

- Click **New Source → Excel Workbook**
- Select **Shipping.xlsx**
- Power BI reads the file and shows available tables, sheets, or named ranges.
- Check the required sheet/table (example: Shipping Table).
- Click **OK** to load it into Power Query.

Power BI extracts the dataset and adds it as a third query.

### 4.2. Verifying Column Headers and Data Types

After loading, inspect each field:

- **Text fields**
  City, Country, Customer ID, Order ID, Product Name, etc.
- **Numeric fields**
  Quantity, Profit, Postal Code
- **Date fields**
  Order Date, Ship Date
- **Categorical fields**
  Ship Mode (Standard Class, Express, etc.)

Power Query may automatically detect correct types, but it is important to confirm or manually adjust them using:

**Transform → Data Type**

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## 5. Loading Data into Power BI Desktop

Once inspection is complete:

- Click **Close & Apply**.
- Power BI begins processing:
  - Connecting to source files
  - Validating queries
  - Loading data into the model

This step may take several seconds depending on file size.

Power BI then displays:

- Tables in the **Fields pane**
- Data in **Table View**
- Visual environment in **Report View**

## 6. Exploring Loaded Tables

Now, in Power BI Desktop:

### 6.1. Appended Table

This contains combined data from the CSVs.

### 6.2. 2018–2019 CSV (not needed now)

Original file (kept for reference).

### 6.3. 2020–2021 CSV (not needed now)

### 6.4. Shipping Table (from Excel)

However, since the appended table contains all order information, we only use:

- **Appended Table**
- **Shipping Table**

The original CSV tables remain unused (but can be hidden or deleted).

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## 7. Moving to Model View for Relationship Building

Data modeling is one of the most critical steps. It ensures tables interact correctly during analysis.

### 7.1. Opening the Model View

Click the **Model** icon on the left pane.

You will see boxes representing each table.

### 7.2. Identifying Relationship Keys

Both **Appended Table** and **Shipping Table** contain:

- Order ID
- Customer ID
- Product ID
- Order Date
- Region, Country
- Category/Sub-Category

Among these, **Order ID** is the most reliable primary key.

### 7.3. Creating the Relationship

To link the tables:

- Drag **Order ID** from the Appended Table
- Drop it onto **Order ID** in the Shipping Table

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Power BI automatically detects:

- **Cardinality** → Many-to-Many
- **Cross Filter Direction** → Both
- **Active Relationship** → ON

Click **OK** to confirm.

### 7.4. Understanding Why Relationship Is Needed

Without a relationship:

- Filters applied to the Appended Table would not impact Shipping Table visuals.
- Measures computed from one table cannot be correctly aggregated with fields from the other.

Thus, the join ensures all KPIs and visuals behave correctly.

## 8. Switching to Report View

Once the relationship is established:

- Go back to **Report View**
- Now we can begin building the dashboard including KPIs.

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## 9. Creating the Year Slicer (Filter for 2021 Reports)

Since the dashboard should display data for the year 2021, we first add a slicer.

### 9.1. Adding a Slicer Visual

- Click **Slicer** from Visualizations.
- A blank slicer appears.

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### 9.2. Adding Year Field

In Appended Table:

- Expand **Order Date**
- Use the **Year** hierarchy field
- Drag and place it in the slicer

Power BI automatically generates a:

- Range slider
- Or dropdown (after formatting)

### 9.3. Formatting the Slicer

Open **Format Visual**:

**Slicer Settings**

- Change **Style**:
  From *Between* → *Dropdown* or *Tile*
- Enable **Single Select**
- Disable **Select All**

**Slicer Header**

- Increase font size (e.g., 20)

**Visual Border**

- Turn on
- Add corner rounding (e.g., 15)

**General Settings**

- Adjust the slicer size and place it near the title/logo.

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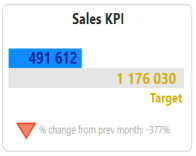
The slicer is now ready and will control the entire report.

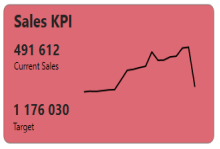
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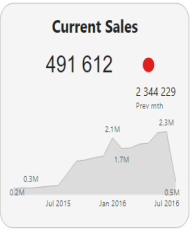
## 10. Creating KPI Cards

The KPIs include:

- Total Orders
- Total Sales
- Total Profit
- Total Quantity
- Average Discount
- Profitability
- Quantity shipped using Standard Class for Accessories




### 10.1. Creating the "Total Orders" KPI

Steps:

1. Insert a **Card** visual.
2. Drag **Order ID** into the card.
3. Instead of showing the order number, change aggregation:
    - Click the dropdown → **Count Distinct**

Using **Distinct Count** ensures:

- Each unique order is counted only once.
- Repeated order IDs (due to multiple items) are not double-counted.

Example:

- Distinct Count → 1687 orders
- Count → 3312 records (misleading because each item in the same order is counted)

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## 10.2. Formatting the Card

**Remove Category Label**

- Remove “Count of Order ID” text below the number.

**Callout Value**

- Increase font size so the number stands out.

**Add Title**

In **General → Title**:

- Turn ON
- Enter: **Total Orders**
- Set font size (e.g., 25)
- Center-align the title

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**Add Borders**

Under **Effects → Visual Border**:

- Turn ON
- Corner rounding = 15

Resize and place it in the layout.

Repeat similar steps for other KPI cards (Sales, Profit, Quantity, Discount).

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## 11. Importance of Distinct Count vs Count

Understanding this choice is crucial:

### Distinct Count

Counts each order ID once
→ Ideal for KPI "Total Orders"

### Count

Counts every row
→ Useful for:

- Number of items ordered
- Number of shipping records
- Quantity lines, etc.

Using the wrong aggregation can mislead business teams.

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## 12. Final Checks Before Moving Forward

Before building remaining visuals:

- Ensure year slicer affects all visuals.
- Confirm data model relationship is active.
- Validate that Appended Table correctly includes all years.
- Validate Shipping Table is connected using Order ID.
- Arrange KPIs neatly on the report canvas.

This completes the foundational setup required for further report creation.

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## 13.Conclusion

This detailed walkthrough covered:

- Importing multiple CSV files with identical structures
- Appending them into a single fact table
- Importing and validating an Excel dataset
- Understanding data types
- Loading data into Power BI Desktop
- Establishing relationships using Model View
- Creating slicers for year-based filtering
- Designing KPI cards with correct aggregations
- Formatting visuals for a clean UI

This is the foundation upon which the full Serveto Sales & Shipping dashboard will be built.

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